

Brain Tumor Detection Using Deep Learning Techniques

Abdullah Dahabre

Abdulrahman Zahir

Abstract

Brain tumors require early and accurate detection to improve patient outcomes and enable timely treatment. Manual analysis of MRI scans is often time-consuming and prone to inconsistencies. This project evaluates several pretrained convolutional neural networks (CNNs) for brain tumor detection using a dataset of 253 labeled MRI images. The images were preprocessed by cropping, resizing, and augmented to improve model performance. The dataset was split into training, validation, and testing sets with stratified sampling to maintain class balance. Model performance was assessed using accuracy, precision, recall, and F1-score. Results show that transfer learning with pre-trained CNNs can classify brain tumor presence effectively despite the limited dataset size. Challenges such as data scarcity, class imbalance, and overfitting were addressed using augmentation and regularization techniques. This project highlights the potential of deep learning models to support clinical diagnosis of brain tumors.

1. Introduction

Medical Problem

Brain tumors pose a significant medical challenge due to their potential to cause severe neurological damage and death if not detected early. Accurate and timely diagnosis is essential for effective treatment planning and improving patient survival rates. Currently, diagnosis depends on manual review of Magnetic Resonance Imaging (MRI) scans by radiologists, which can be time-consuming and prone to variability and human error.

Importance in Medical/Clinical Context

Early and accurate detection of brain tumors is crucial to reduce diagnostic delays and improve consistency and reliability. Timely diagnosis enables faster intervention, leading to better treatment outcomes and enhanced patient quality of life.

Objectives and Scope

This project evaluates the performance of pretrained convolutional neural networks for classifying brain MRI images into tumor and non-tumor categories. The work involves preprocessing MRI data, applying transfer learning with multiple CNN architectures, and comparing their effectiveness using key evaluation metrics. The goal is to explore the potential of deep learning models to assist clinical decision-making in brain tumor diagnosis.

2. Background and Literature Review

Overview of Related Work

Sharma et al. focused on brain tumor detection using traditional machine learning applied to MRI images[1]. Their approach involved automated preprocessing steps including grayscale conversion, noise removal with median filtering, edge detection, and watershed segmentation to isolate tumor regions. Features were automatically extracted using the Gray Level Co-occurrence Matrix (GLCM), capturing texture properties such as energy, contrast, correlation, and homogeneity. These handcrafted features were then classified using algorithms like Multilayer Perceptron (MLP) and Naive Bayes.

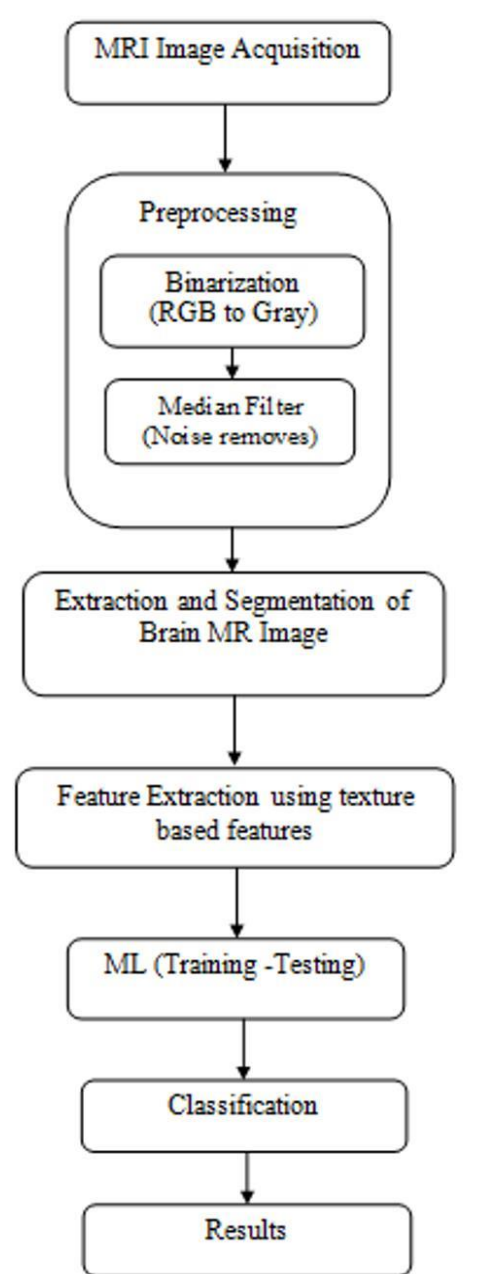


Figure 1. Automated MRI-Based Brain Tumor Classification Pipeline

Common Challenges and Existing Solutions

Common challenges include variability in tumor shape, size, and location, as well as noise in MRI scans. Manual feature extraction can be labor-intensive and may miss subtle patterns, but Sharma et al. addressed this by automating preprocessing and segmentation to improve feature quality. However, limited dataset size and potential overfitting remain ongoing issues in machine learning-based brain tumor detection.

3. Dataset Description

Source and Nature

The dataset used is a publicly available collection of brain MRI images from Kaggle[2], containing 253 labeled scans with 155 tumor-positive and 98 tumor-negative cases. The images are MRI scans focused on brain regions for binary classification of tumor presence.

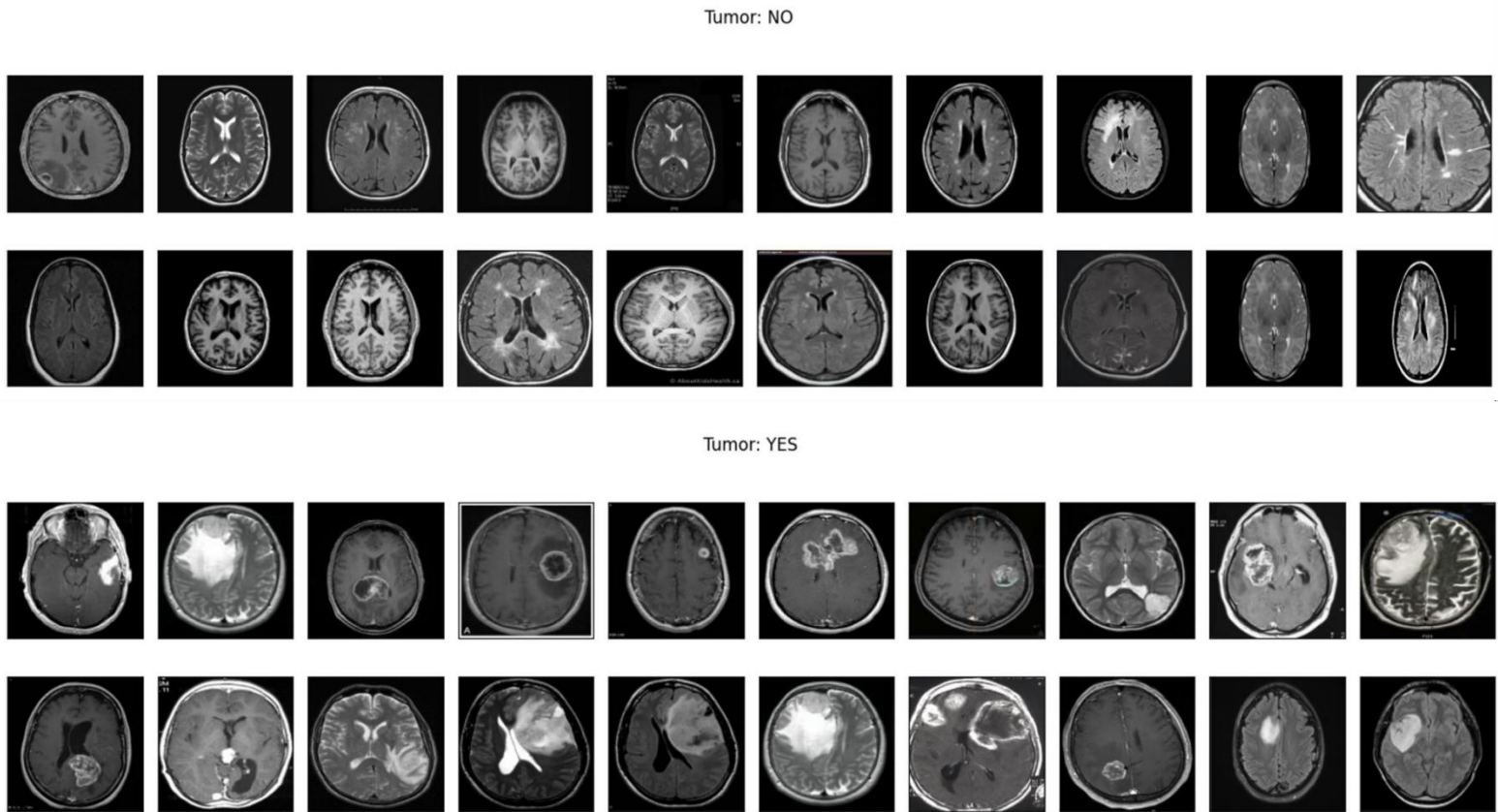


Figure 2. Sample Images from Our Brain MRI Dataset (Non-Tumor Top Rows; Tumor Bottom Rows)

Preprocessing Steps

Both Abdullah and Abdulrahman based their work on the publicly available Kaggle notebook by Ruslan Kl. (2023)[3], adapting and extending the original code to suit their experiments. We both applied similar preprocessing steps including cropping the brain region using contour detection and resizing images to 224×224 pixels to meet the input size requirements of pre-trained CNNs. Normalization of pixel values was performed in both notebooks to standardize the input.

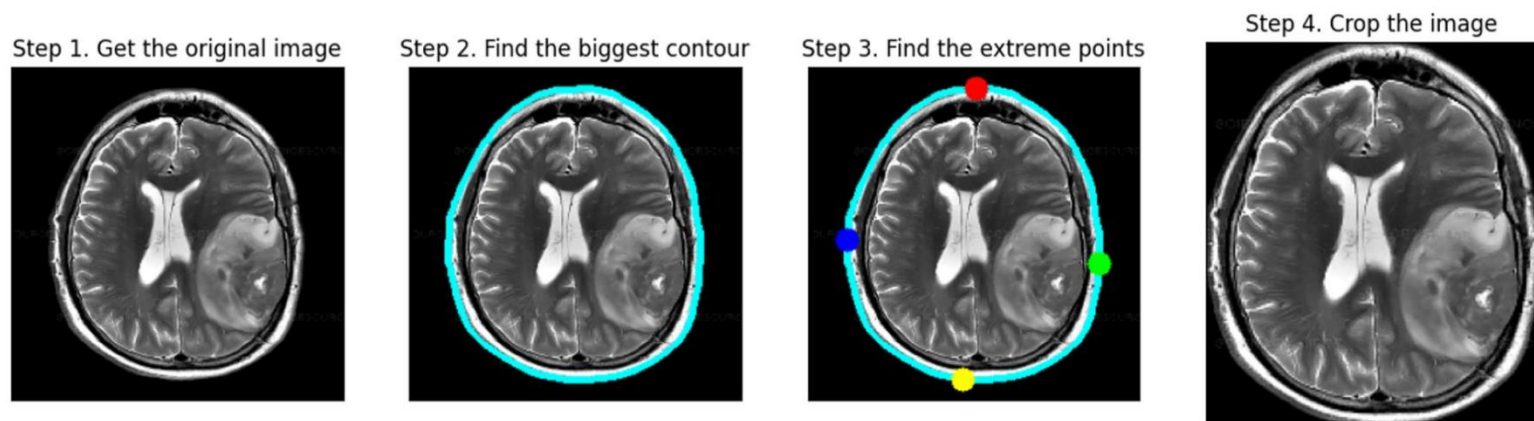


Figure 3. Brain MRI Preprocessing Workflow: Cropping the Brain Region

For data augmentation, both implemented common transformations such as rotation, zoom, horizontal flipping, and width/height shifting to enhance dataset variability and reduce overfitting risk.

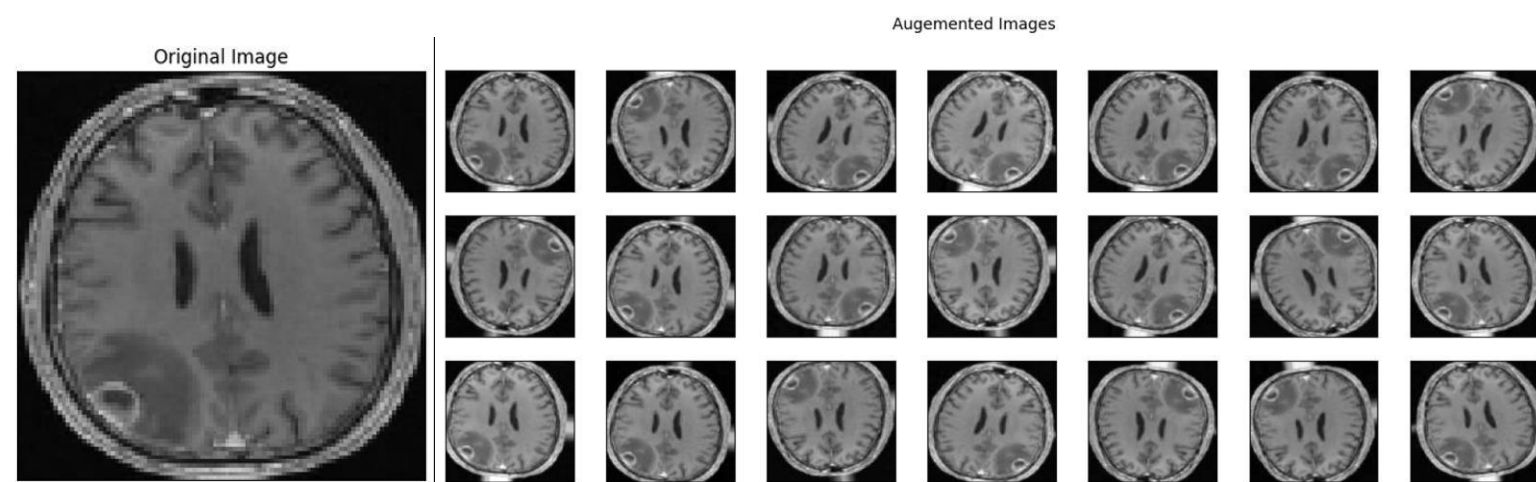


Figure 4. Original MRI Slice and Example Augmented Variations

Data Partitioning

We both used stratified splitting to maintain class balance in train, validation, and test sets. The common split ratio was 80% training, 10% validation, and 10% testing.



Figure 5. Class Distribution Across Training, Validation, and Test Sets

- Abdullah used a random seed of 42 during splitting, with two-step train_test_split calls: first separating test data, then splitting the remaining into train and validation.
- Abdulrahman applied a random seed of 123 and slightly different proportions in the second split (using a test size of 0.11 for validation, approximating 10% of the total).

Despite these minor differences, both approaches ensured consistent and balanced splits to fairly evaluate model performance.

4. Methodology

Choice of Models

Both Abdullah and Abdulrahman leveraged convolutional neural networks (CNNs) with transfer learning for brain tumor detection. Abdullah focused on VGG16, VGG19, and InceptionV3, while Abdulrahman experimented with alternative pre-trained architectures such as ResNet50, EfficientNetB0, and MobileNetV2. Transfer learning enabled us to benefit from features learned on large datasets like ImageNet, improving performance despite limited MRI data.

Architecture Details, Loss Functions and Optimization and Hyperparameters

Aspect	VGG16	VGG19	InceptionV3	ResNet50	EfficientNet B0	MobileNetV2
Architecture Details	VGG16 base (frozen)→ Flatten→ Dropout → Dense (Sigmoid)	VGG19 base (frozen)→ Flatten→ Dense(256, ReLU + L2)→ Dropout→ Dense(Sigmoid)	InceptionV3 base(unfrozen last 30 layers)→ GAP→ Dropout→ Dense(128, ReLU)→ Dropout→ Dense(Sigmoid)	ResNet50 base (frozen)→ GAP→ Dropout→ Dense(128)→ Sigmoid	EfficientNetB0 base (frozen)→ GAP→ Dropout→ Dense layers→ Dropout→ Sigmoid	(Frozen)→ GAP→ Dropout→ Dense layers→ Dropout→ Sigmoid
Loss Function	Binary Crossentropy	Focal Loss($\gamma=2.0$, $\alpha=0.25$)	Focal Loss($\gamma=2.0$, $\alpha=0.25$)	Binary Crossentropy	Binary Crossentropy	Binary Crossentropy
Optimizer	RMSProp	Adam	Adam	Adam	Adam	Adam

Hyperparameters	LR = 1e-4 Batch = 32 Epochs = 14(stopped early)	LR = 1e-4(with ReduceLROnPlateau) Batch = 32 Epochs = 18(stopped early)	LR = 1e-4(with ReduceLROnPlateau) Batch = 32 Epochs = 10 (frozen) + 26(fine-tuning) = 36	LR = 1e-4 Batch = 32 Epochs = 30	LR = 1e-4 Batch = 32 Epochs = 30	LR = 1e-4 Batch = 32 Epochs = 30
Regularization	Dropout (0.5)	Dropout (0.5) + L2 (0.001)	Dropout (0.5, 0.3)	Dropout (0.5)	Dropout (0.5, 0.3)	Dropout (0.5, 0.3)
Class Weighting	Not used	Not used	Balanced class weights	Balanced class weights	Balanced class weights	Balanced class weights
Pretrained Weights	ImageNet	ImageNet	ImageNet	ImageNet	ImageNet	ImageNet

Tools and Libraries

We both utilized Python with TensorFlow and Keras frameworks for model implementation and training. Additional libraries included NumPy, OpenCV for image preprocessing, and scikit-learn for data splitting and evaluation metrics. Data augmentation was implemented using Keras’ ImageDataGenerator.

5. Evaluation Metrics

Metrics Selection

Multiple evaluation metrics were selected to provide a comprehensive assessment of model performance:

- 1. **Accuracy:** Proportion of correctly classified images (both tumor and non-tumor)
- 2. **Precision:** Proportion of true positive predictions among all positive predictions (tumor)
- 3. **Recall (Sensitivity):** Proportion of true positive predictions among all actual positive cases
- 4. **F1-Score:** Harmonic mean of precision and recall
- 5. **Confusion Matrix:** Visual representation of prediction errors and correct classifications

Justification for Metric Choice

In medical diagnostic contexts, both false positives and false negatives have significant implications:

- **False Positives** (incorrectly identifying a healthy brain as having a tumor) can lead to unnecessary anxiety, additional testing, and potentially invasive procedures.

- **False Negatives** (missing an actual tumor) can delay critical treatment and negatively impact patient outcomes.

Therefore, a balanced consideration of precision and recall is essential. The F1-score provides this balance by combining both metrics. Additionally, accuracy offers an overall performance measure, while the confusion matrix provides detailed insight into the types of errors made by each model.

6. Results

Quantitative Results

Performance metrics for all six models on the **test set**:

Aspect	VGG16	VGG19	InceptionV3	ResNet50	EfficientNetB0	MobileNetV2
Accuracy	0.89	0.83	0.94	0.88	0.88	0.80
Precision	0.91	0.88	0.95	0.89	0.90	0.85
Recall	0.89	0.83	0.94	0.88	0.88	0.80
F1-Score	0.89	0.83	0.94	0.88	0.88	0.78

The InceptionV3 model achieved the highest performance across all metrics, with 94% accuracy, 95% precision, and 94% recall. VGG16, ResNet50, and EfficientNetB0 showed comparable performance (88-89% accuracy), while MobileNetV2 had the lowest performance at 80% accuracy.

Visual Results

Confusion matrix of best model (test set):

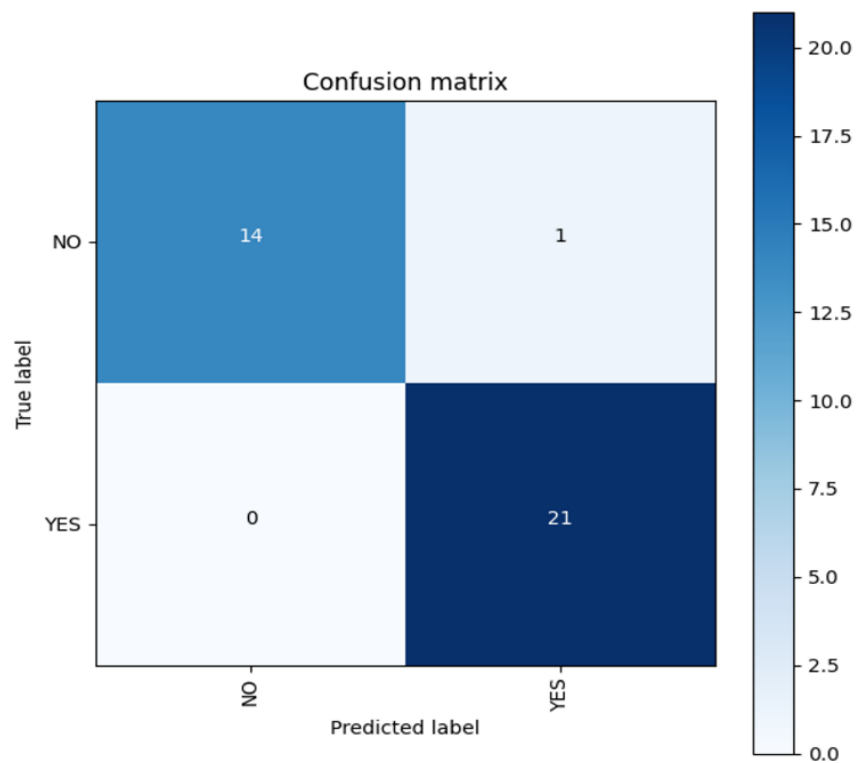


Figure 6. Confusion matrix of InceptionV3 on test set

InceptionV3 achieved near-perfect classification, correctly identifying all tumor cases and misclassifying only one non-tumor case.

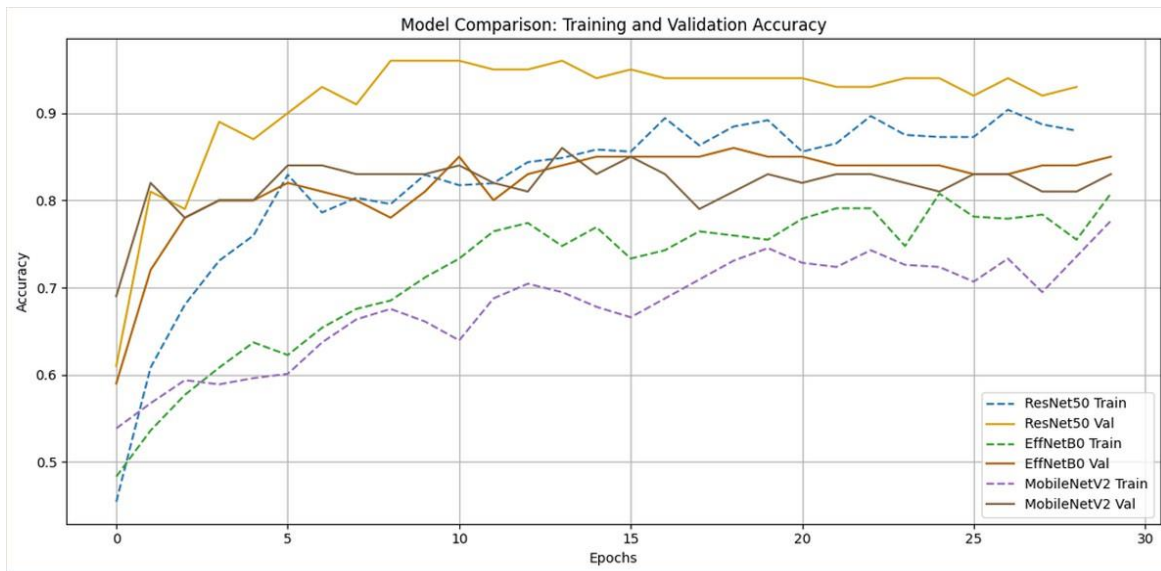


Figure 7. Training and validation accuracy for VGG16, VGG19, and InceptionV3

The first plot compares the training and validation accuracy of three pretrained CNN models: VGG16, VGG19, and InceptionV3 over multiple epochs. VGG19 and VGG16 show relatively stable validation accuracy around 90%, while InceptionV3 achieves the highest training accuracy but has lower validation accuracy, indicating some overfitting.

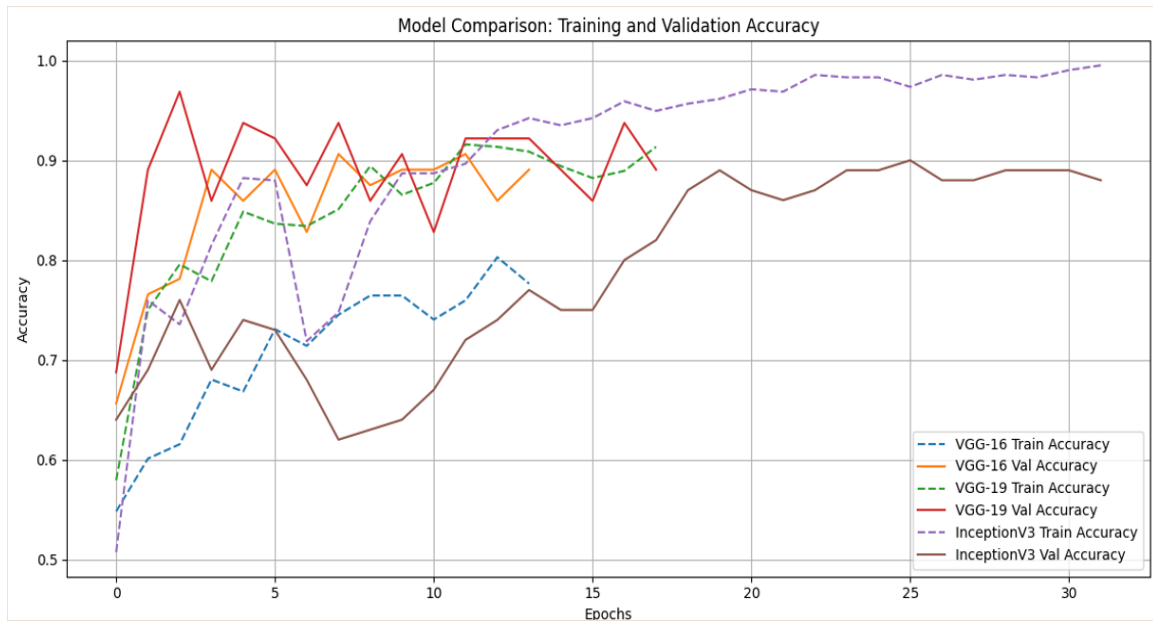


Figure 8. Training and validation accuracy for ResNet50, EfficientNetB0, and MobileNetV2

The second plot shows that ResNet50 achieves the highest validation accuracy nearing 95%, outperforming the other models. EfficientNetB0 and MobileNetV2 exhibit moderate performance with validation accuracy stabilizing around 80–85%. The plots suggest ResNet50 has better generalization on this dataset.

Comparison with Literature

ML Algorithm	Total samples	Model Build Time	Classification Rate (%)
MLP	210	61.82	98.6
Naive bayes	210	0.02	97.6

Figure 9. Experimental result analysis of the paper

Sharma et al. reported that the Multilayer Perceptron (MLP) achieved higher accuracy (98.6%) but required longer training time, while Naive Bayes trained much faster with slightly lower accuracy (97.6%), illustrating the trade-off between accuracy and efficiency .

Sharma et al. used handcrafted texture features extracted via the Gray Level Co-occurrence Matrix (GLCM) and trained classical classifiers—Multilayer Perceptron (MLP) and Naive Bayes—on a similar MRI dataset, achieving accuracies above 97% with minimal data preprocessing .

In contrast, our approach employed six pre-trained CNN architectures (VGG16, VGG19, InceptionV3, ResNet50, EfficientNetB0, MobileNetV2) on 253 images. Despite their powerful feature-learning

capabilities, these deep models yielded lower overall accuracy (~90%) due to the limited dataset size and increased overfitting risk.

This comparison underscores that, when data are scarce, simpler classical ML methods with handcrafted features can sometimes outperform complex deep learning models.

7. Discussion

Interpretation of Results

The pretrained CNN models demonstrated the capability to classify brain MRI images into tumor and non-tumor categories, although with moderate accuracy compared to some classical approaches in the literature. The use of transfer learning enabled the models to extract meaningful features despite the small dataset. However, the results suggest that some models struggled with overfitting and class imbalance, which affected generalization to unseen data.

Limitations

A major limitation was the relatively small size of the dataset (253 images), which is insufficient for fully training deep neural networks without overfitting. The class imbalance—with more tumor images than non-tumor—also likely skewed training, despite stratified splitting and data augmentation efforts. Another limitation is the use of a single data source, which may not fully represent the diversity of MRI images encountered in clinical practice, limiting the generalizability of the model.

Sources of Error

Potential sources of error include noise and variability inherent in MRI scans, imperfect segmentation of the brain region during preprocessing, and limitations in augmentation techniques that may not fully capture the diversity of real tumor presentations. The limited number of samples also restricted the models' exposure to rare tumor variations, impacting predictive accuracy.

Clinical Relevance and Impact

Despite moderate performance, this study demonstrates that pre-trained CNN models can support brain tumor classification tasks, providing a foundation for AI-assisted diagnostic tools. With larger and more balanced datasets, these models could help reduce diagnostic time and variability, assisting radiologists in early detection and treatment planning. Further improvements may enhance clinical applicability and patient outcomes.

References

1. **Sharma, K., Kaur, A., & Gujral, S. (n.d.).** Brain tumor detection based on machine learning algorithms. Chandigarh University, Gharuan, Punjab, India.
2. **Chakrabarty, N. (n.d.).** Brain MRI images for brain tumor detection Dataset. Kaggle. <https://www.kaggle.com/datasets/navoneel/brain-mri-images-for-brain-tumor-detection>
3. **Klyukin, R. (2019).** Brain tumor detection v1.0 (CNN + VGG-16). Kaggle. <https://www.kaggle.com/code/ruslankl/brain-tumor-detection-v1-0-cnn-vgg-16/notebook>
4. **Chollet, F. (2015).** Keras. GitHub. <https://github.com/fchollet/keras>

5. **Simonyan, K., & Zisserman, A. (2015).** Very deep convolutional networks for large-scale image recognition. International Conference on Learning Representations (ICLR). <https://arxiv.org/abs/1409.1556>
6. **Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., & Wojna, Z. (2016).** Rethinking the inception architecture for computer vision. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2818-2826. <https://arxiv.org/abs/1512.00567>
7. **He, K., Zhang, X., Ren, S., & Sun, J. (2016).** Deep residual learning for image recognition. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 770-778. <https://arxiv.org/abs/1512.03385>
8. **Tan, M., & Le, Q. V. (2019).** EfficientNet: Rethinking model scaling for convolutional neural networks. Proceedings of the 36th International Conference on Machine Learning (ICML), 6105-6114. <https://arxiv.org/abs/1905.11946>
9. **Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., & Chen, L. C. (2018).** MobileNetV2: Inverted residuals and linear bottlenecks. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 4510-4520. <https://arxiv.org/abs/1801.04381>
10. **Kingma, D. P., & Ba, J. (2015).** Adam: A method for stochastic optimization. International Conference on Learning Representations (ICLR). <https://arxiv.org/abs/1412.6980>
11. **Lin, T. Y., Goyal, P., Girshick, R., He, K., & Dollár, P. (2017).** Focal loss for dense object detection. Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2980-2988. <https://arxiv.org/abs/1708.02002>
12. **Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., van der Laak, J. A. W. M., van Ginneken, B., & Sánchez, C. I. (2017).** A survey on deep learning in medical image analysis. Medical Image Analysis, 42, 60-88. <https://doi.org/10.1016/j.media.2017.07.005>